ZINEDDINE REBBOUH

Full Stack Engineer · AI/ML Engineer · Data Scientist

+213 562 163 159 · zinedinerabouh@gmail.com · Touggourt, Algeria · github.com/Zineddine-Rebbouh · zinedine-rebbouh-portfoliowebsite.vercel.app · Immediately available — open to relocation

PROFESSIONAL SUMMARY

Full Stack Engineer and AI/ML practitioner with 2+ years of end-to-end product development experience, currently completing an M.Sc. in Artificial Intelligence & Data Science. Proven ability to architect and ship production-grade systems from concept to cloud deployment: REST API design, real-time WebSocket pipelines, async task queues (Celery + Redis), and ML model integration. Author of **Tawfir**, Algeria's first AI-powered food waste redistribution platform, deployed on AWS with integrated ML/NLP pipelines. Ranked **1st** in undergraduate thesis competition (17/20). Delivered deep learning models exceeding state-of-the-art benchmarks (3D CNN: 90.71% accuracy, +75% vs prior LSTM; DeepLabV3+ segmentation: IoU 0.8001, within 2.3% of competition winner). English C1 certified. Immediately available for full-time on-site or remote roles.

TECHNICAL SKILLS

Languages:	Python (Advanced), JavaScript/TypeScript (Advanced), SQL, Java
Frontend:	React.js, Next.js (SSR/SSG/App Router), TypeScript, Redux Toolkit, Tailwind CSS, Material UI, Bootstrap, HTML5/CSS3, Flutter (cross-platform)
Backend:	Django 5.0, Django REST Framework, Node.js, Express.js — REST API design, JWT/OAuth2, RBAC, middleware, async processing
Real-Time:	Django Channels (WebSocket), Redis pub/sub, Celery + django-celery-beat — event-driven & scheduled async architectures
Databases:	PostgreSQL 15 + PostGIS, MongoDB, MySQL, Redis — schema design, indexing, query optimization, geospatial queries
AI / ML / DL:	TensorFlow 2.x/Keras, PyTorch 2.0, Scikit-learn, XGBoost, HuggingFace Transformers — CNN, 3D CNN, U-Net, DeepLabV3+, NLP, time-series forecasting (ARIMA/Prophet), recommender systems
Computer Vision:	Image segmentation (semantic/binary), video classification, MediaPipe, CLAHE, OpenCV, data augmentation pipelines
Data & Analysis:	Pandas, NumPy, Matplotlib, Seaborn, SciPy — EDA, feature engineering, SMOTE, evaluation metrics
MLOps / Cloud:	AWS (EC2, RDS, S3, CloudFront), Docker, Git/GitHub, CI/CD (GitHub Actions), Kaggle GPU (Tesla P100/T4), Linux/Ubuntu, nginx
Testing & Quality:	Playwright (E2E), Jest, React Testing Library, Postman — well-tested, documented software delivery
Architecture:	SOLID principles, clean code, modular design, microservices, DRY, event-driven systems

PROFESSIONAL EXPERIENCE

Machine Learning Engineer — Elevvo Pathways

Jul 2025 – Aug 2025

Remote · Cairo, Egypt

- Designed and delivered 6 complete end-to-end ML pipelines across regression, classification, clustering, forecasting, recommender systems, and computer vision — each with production-quality Python code, rigorous evaluation metrics, and structured English technical documentation.
- Regression: Student exam score prediction using Linear & Ridge Regression with feature correlation analysis, missing value imputation, RMSE/R^2 evaluation.
- Classification: Multi-class forest cover type (Random Forest, SVM) and binary loan approval model with SMOTE oversampling; reported precision, recall, F1 per class with confusion matrix analysis.
- Clustering: K-Means customer segmentation with elbow method and silhouette scoring; produced business-ready segment profile visualisations.
- Forecasting: XGBoost Walmart weekly sales pipeline with lag features, rolling statistics, and GridSearchCV hyperparameter optimisation — competitive MAE on held-out test set.
- Recommender System: Collaborative filtering movie recommendation engine via matrix factorisation; evaluated with precision@k and recall@k.
- Computer Vision: CNN traffic sign recognition with data augmentation pipeline and filter activation analysis.
- All 6 projects delivered within 2-month timeline in 100% remote international team — proficient with collaborative Git workflows and async communication.

React.js Frontend Developer — Codsoft

Mar 2024 – Apr 2024

Remote

- Designed, built, and shipped 3 production-grade React SPAs (To-Do App, Portfolio, Blog Platform) within a single 4-week sprint — complete with reusable component architecture, responsive Tailwind CSS UI, and optimised state management.
- Built shared component libraries applying separation of concerns and DRY principles — reducing code duplication and accelerating feature delivery.
- Applied strict performance patterns (useMemo/useCallback) to eliminate unnecessary re-renders; ensured full cross-browser and mobile responsiveness on all three applications.
- Identified and resolved anti-patterns: state mutation bugs, improper useEffect dependency arrays, and uncontrolled side effects — demonstrating strong code-quality ownership.
- Integrated REST APIs into frontend applications with full async handling, loading states, and error boundaries.

KEY PROJECTS

Tawfir — AI-Powered Food Waste Redistribution Platform (M.Sc. Thesis) 2024 – Present

Stack: Flutter · Django 5.0 · Django REST Framework · PostgreSQL 15 + PostGIS · Redis · Celery · Docker · AWS (EC2, RDS, S3, CloudFront) · Stripe · BaridiMob · Firebase · ARIMA/Prophet · HuggingFace · Scikit-learn

- Architected and built Algeria's first AI-driven food waste redistribution marketplace from scratch — a three-sided platform connecting consumers, merchants, and charitable associations to address 10M+ tons of annual food waste.
- Backend: Django REST Framework 5.0 with 40+ RESTful endpoints, JWT/OAuth2 authentication, role-based access control (RBAC), geospatial queries via PostGIS, Redis caching + pub/sub message broker, Celery + django-celery-beat for scheduled async task pipelines, and full Docker containerisation.
- Real-Time: Django Channels WebSocket for live order tracking and push notifications; Firebase Cloud Messaging for mobile push alerts.
- Frontend/Mobile: Cross-platform Flutter application (iOS, Android, Web) with full state management, QR code authentication, deep linking, and offline-capable caching layer. React web admin dashboard with role-specific views.
- AI/ML Components: ARIMA & Prophet time-series models for surplus prediction and demand forecasting; collaborative filtering recommendation engine matching users to nearby surplus offers; NLP sentiment analysis pipeline for user review processing — all served via clean REST API interfaces.
- Payments: Dual payment gateway integration — Stripe (international cards) and BaridiMob (Algerian CCP), with complete transaction lifecycle management, refunds, and webhook handling.
- Cloud & DevOps: Deployed on AWS (EC2 for application server, RDS for PostgreSQL, S3 for media storage, CloudFront CDN); CI/CD pipeline with GitHub Actions; nginx reverse proxy; SSL/TLS configured; production monitoring setup.
- Data: PostgreSQL 15 with PostGIS for geospatial product/location queries; Redis for caching high-read endpoints and WebSocket message brokering; S3 for media assets.

Arabic Sign Language Detection — Real-Time Recognition System (M.Sc. Mini-Project) 2025 – 2026

Stack: PyTorch 2.0 · MediaPipe · OpenCV · Albumentations · Scikit-learn · Kaggle T4 GPU

- Built end-to-end Arabic sign language recognition system: custom 3D CNN (14.3M parameters, 4 Conv3D blocks + AdaptiveAvgPool + 3-layer classifier) achieving 90.71% validation accuracy on a 20-class video dataset — vs 15.75% prior LSTM baseline on the same dataset (75% improvement).
- Engineered comprehensive preprocessing pipeline: MediaPipe hand detection (21 keypoints, 0.3 confidence threshold) → ROI extraction with 50% padding → CLAHE contrast enhancement → normalisation to [-1,1]; hand detection alone contributed +19.51% accuracy gain (documented ablation study).
- Overfitting mitigation: Dropout (0.25–0.5), label smoothing (0.1), cosine annealing LR schedule, gradient clipping (1.0), AdamW optimiser — reduced train/val accuracy gap from 15% to 8.3%.
- Developed real-time webcam inference application with live confidence display; full technical documentation with confusion matrix and per-class error analysis.

Image Segmentation — Classical vs Deep Learning Benchmark (M.Sc. Mini-Project) 2025 – 2026

Stack: TensorFlow 2.x / Keras · OpenCV · NumPy · SciPy · Kaggle Tesla P100 GPU

- Benchmarked 6 segmentation architectures under identical conditions: U-Net, U-Net++, DeepLabV3+ (deep learning) vs Otsu, K-Means, Watershed (classical) — across urban scene (CamVid) and medical imaging (ISIC 2018) datasets.
- DeepLabV3+ achieved IoU 0.8001 on CamVid and Dice 0.8863 on ISIC 2018 — within 2.3% of ISIC 2018 competition winner using a single model vs ensemble approaches.
- Phased transfer learning (frozen ResNet-50 warm-start → full fine-tuning) achieved 2× faster convergence vs training from scratch; quantified 29.5% absolute IoU gap between best classical (K-Means: 0.62) and best DL method with documented failure mode root-cause analysis.

WANLP — Real-Time NLP Sports Trend Detection Platform Feb 2025 – Apr 2025

Stack: Next.js · Django Channels · Redis · HuggingFace Transformers · Stanza NER · Celery · MongoDB

- Architected full-stack NLP platform processing 1,000+ social media events/minute: Stanza NER + fine-tuned HuggingFace Transformer for multi-class sentiment classification → Django Channels + Redis WebSocket → Next.js TypeScript live dashboard.

- Async data pipeline: Twitter scraping (Twikit) → NER → Sentiment Analysis → MongoDB aggregation → real-time ranked trend feed; Celery + django-celery-beat for scheduled background ingestion; sub-second live frontend updates with zero polling overhead.
- Optimised MongoDB aggregation queries and Redis caching for low-latency trend lookups under concurrent user load; Next.js SSR frontend deployed to Vercel.

HotelPro — Full-Stack Hotel Management System 2025

Stack: React + TypeScript + Vite · Node.js + Express · MongoDB · Stripe · Playwright

- Complete MERN application: React/TypeScript/Vite frontend with strict TypeScript coverage on all component props and API shapes; Node.js/Express backend, MongoDB, JWT authentication, Stripe payments.
- Multi-role dashboard (admin, staff, guest) with Redux Toolkit — complex state architecture serving three distinct user journeys from a single SPA.
- End-to-end Playwright test suite covering critical user journeys (booking, authentication, admin panel); comprehensive documentation covering state flows and API contracts.

Multi-Vendor E-Commerce Platform (B.Sc. Graduation Project — 1st Place) Feb 2024 – May 2024

Stack: React.js · Node.js · Express.js · MongoDB Atlas · JWT · Vercel

- Built complete MERN-stack multivendor marketplace supporting multi-vendor operations, admin controls, and customer transactions — 20+ RESTful endpoints, JWT authentication with RBAC enforced at middleware level.
- Designed MongoDB schemas with strategic indexing optimised for high-read product catalog performance; implemented pagination and query optimisation for scalability.
- Identified and resolved 12+ production bugs including auth token edge cases, concurrent cart conflicts, and async error propagation — documented root causes and fixes.
- Ranked 1st in B.Sc. thesis competition (17/20) — defended in English before faculty jury.

EDUCATION

M.Sc. Artificial Intelligence & Data Science 2024 – 2026 (Near completion)

University of Constantine 2 – Abdelhamid Mehri, Algeria

- Thesis: Tawfir — production AI platform deployed on AWS with integrated ML/NLP pipelines.
- Mini-projects: Image Segmentation (DeepLabV3+, IoU 0.8001) · Arabic Sign Language Detection (3D CNN, 90.71% accuracy).
- Coursework: Statistical Learning, Deep Learning, NLP, Optimisation for ML, Data Mining, Big Data Systems.

B.Sc. Computer Science 2021 – 2024

University of Constantine 2 – Abdelhamid Mehri, Algeria

- Ranked 1st in undergraduate thesis competition — score 17/20 across the entire cohort; defended in English before faculty jury.
- Coursework: Algorithms & Data Structures, Software Engineering, Database Systems, OOP, Web Development, Computer Networks, Operating Systems.
- Invited to International Mathematics Olympiad 2021, Provincial Round — Ministry of National Education.

CERTIFICATIONS

- Supervised Machine Learning — Andrew Ng · DeepLearning.AI / Coursera
- Advanced Learning Algorithms — Andrew Ng · DeepLearning.AI / Coursera
- Unsupervised Learning, Recommenders & Reinforcement Learning — Andrew Ng · Coursera
- Natural Language Processing — OpenHPI / Hasso Plattner Institute
- Understanding Data Engineering — DataCamp
- EF SET English Certificate — C1 Advanced (certified)

HONOURS & AWARDS

- 1st Place — B.Sc. Graduation Thesis, University of Constantine 2 (2024), score 17/20 — top of entire promotion
- Invited — International Mathematics Olympiad 2021, Provincial Round (Ministry of National Education)
- Participation Certificate — Hackathon Batna, Algeria
- Algeria 3.6 Coding Challenge — National-level competitive programming

ADDITIONAL INFORMATION

Languages: Arabic (Native) · English C1 Advanced (EF SET certified) · French B1 Intermediate

Location: Touggourt, Algeria — immediately available for full-time on-site relocation or remote

Availability:	Immediately available — can join within 7 days
Portfolio:	zinedine-rebbouh-portfoliowebsite.vercel.app
GitHub:	github.com/Zineddine-Rebbouh
Work Style:	Remote-native · ownership mindset · async-friendly · startup-pace delivery